

Capítulo 3

Base inversa

Tabla 3-1

c_i	A_i	20	45	0	0	0	x_i	x_i/y_{ij} ($y_{ij} > 0$)
		A_1	A_2	A_3	A_4	A_5		
0	A_3	1	2	1	0	0	40	20
0	A_4	3	3/2	0	1	0	75	50
0	A_5	0	1	0	0	1	15	15
z_j		0	0	0	0	0	$z = 0$	
$c_i - z_j$		20	45	0	0	0		

Tabla 3-2

c_i	A_i	20	45	0	0	0	x_i	x_i/y_{ij} ($y_{ij} > 0$)
		A_1	A_2	A_3	A_4	A_5		
0	A_3	1	0	1	0	-2	10	10
0	A_4	3	0	0	1	-3/2	105/2	35/2
45	A_2	0	1	0	0	1	15	
z_j		0	45	0	0	45	$z = 675$	
$c_i - z_j$		20	0	0	0	-45		

Tabla 3-3

c_i	A_i	20	45	0	0	0	x_i	x_i/y_{ij} ($y_{ij} > 0$)
		A_1	A_2	A_3	A_4	A_5		
20	A_1	1	0	1	0	-2	10	
0	A_4	0	0	-3	1	9/2	45/2	
45	A_2	0	1	0	0	1	15	
z_j		20	45	20	0	5	$z = 875$	
$c_i - z_j$		0	0	-20	0	-5		

A3	A4	A2
1	0	2
0	1	1.5
0	0	1

Inversa		
1	0	-2
0	1	-1.5
0	0	1

A1	A4	A2
1	0	2
3	1	1.5
0	0	1

Inversa		
1	0	-2
-3	1	4.5
0	0	1

Coeficientes de sustitución y costos reducidos

Del desarrollo de factibilidad

$$N = B^{-1}N \Rightarrow Y = B^{-1}N$$

Tabla 3-1

c_i	A_i	20	45	0	0	0	x_i	x_i/y_{ij} ($y_{ij} > 0$)
		A_1	A_2	A_3	A_4	A_5		
0	A_3	1	2	1	0	0	40	20
0	A_4	3	3/2	0	1	0	75	50
0	A_5	0	1	0	0	1	15	15
z_i		0	0	0	0	0	$z = 0$	
$c_i - z_i$		20	45	0	0	0		

Tabla 3-2

c_i	A_i	20	45	0	0	0	x_i	x_i/y_{ij} ($y_{ij} > 0$)
		A_1	A_2	A_3	A_4	A_5		
0	A_3	1	0	1	0	-2	10	10
0	A_4	3	0	0	1	-3/2	105/2	35/2
45	A_2	0	1	0	0	1	15	
z_i		0	45	0	0	45	$z = 675$	
$c_i - z_i$		20	0	0	0	-45		

$$z = z_0 + x_j (c_j - z_j)$$

$$\frac{\partial z}{\partial x_j} = 0 + c_j - z_j$$

coef. de sustitución

$$Ax = b$$

$$Bx_B + Nx_N = b$$

$$Bx_B = b - Nx_N$$

$$\underbrace{B^{-1}B}_{I} x_B = B^{-1}b - \underbrace{B^{-1}N}_{Y} x_N$$

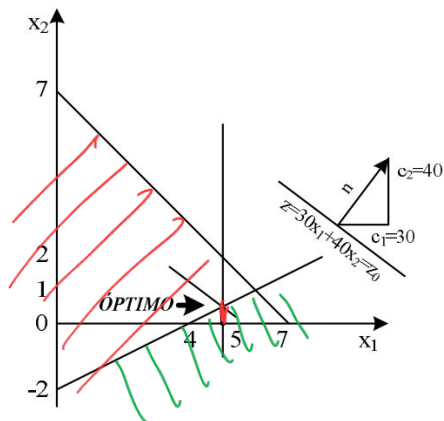
$$x_B = B^{-1}b - Yx_N$$

$$x_B = B^{-1}b - \sum_j Y_j x_j$$

$$\frac{\partial x_B}{\partial x_j} = 0 - \sum_j Y_j = -Y_j$$

$$\frac{\partial x_{B1}}{\partial x_j} = -Y_{1j}; \quad \frac{\partial x_{B2}}{\partial x_j} = -Y_{2j}; \dots$$

Variables ficticias y penalización



Solución Óptima : $x_1^* = 5$, $x_2^* = 1/2$, $z^* = 170$

$$\text{Max } z = 30x_1 + 40x_2$$

s. a

$$x_1 + x_2 \leq 7$$

$$x_1 - 2x_2 \geq 4$$

$$x_1 = 5$$

$$x_1, x_2 \geq 0$$

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4$$

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 = 7$$

$$x_1 - 2x_2 + 0x_3 - 1x_4 = 4$$

$$x_1 + 0x_2 + 0x_3 + 0x_4 = 5$$

$$x_j \geq 0 \quad j = 1, 2, \dots, 4$$

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4 - Mx_5 - Mx_6$$

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 + 0x_5 + 0x_6 = 7$$

$$x_1 - 2x_2 + 0x_3 - 1x_4 + 1x_5 + 0x_6 = 4$$

$$x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 + 1x_6 = 5$$

$$x_j \geq 0 \quad j = 1, 2, \dots, 6$$

HOLGURA / EXCESO

FICTICIAS

Variables ficticias y penalización

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4 - Mx_5 - Mx_6$$

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{0x_6} = 7$$

$$x_1 - 2x_2 + 0x_3 - 1x_4 + \mathbf{1x_5} + \mathbf{0x_6} = 4$$

$$x_1 + 0x_2 + 0x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{1x_6} = 5$$

$$x_j \geq 0 \quad j = 1, 2, \dots, 6$$

c _i	A _i	30	40	0	0	-M	-M	x _i	x _i /y _{ij} (y _{ij} >0)
		A ₁	A ₂	A ₃	A ₄	A ₅	A ₆		
0	A ₃	1	1	1	0	0	0	7	7
-M	A ₅	1	-2	0	-1	1	0	4	4
-M	A ₆	1	0	0	0	0	1	5	5
z _j		-2M	2M	0	M	-M	-M	z = -9M	
c _j - z _j		30+2M	40-2M	0	-M	0	0		

c _i	A _i	30	40	0	0	-M	-M	x _i	x _i /y _{ij} (y _{ij} >0)
		A ₁	A ₂	A ₃	A ₄	A ₅	A ₆		
0	A ₃	0	3	1	1	-1	0	3	1
30	A ₁	1	-2	0	-1	1	0	4	
-M	A ₆	0	2	0	1	-1	1	1	1/2
z _j		30	-60-2M	0	-30-M	30+M	-M	z = 120-M	
c _j - z _j		0	100+2M	0	30+M	-30-2M	0		

c _i	A _i	30	40	0	0	-M	-M	x _i	x _i /y _{ij} (y _{ij} >0)
		A ₁	A ₂	A ₃	A ₄	A ₅	A ₆		
0	A ₃	0	0	1	-1/2	1/2	-3/2	3/2	
30	A ₁	1	0	0	0	0	1	5	
40	A ₂	0	1	0	1/2	-1/2	1/2	1/2	
z _j		30	40	0	20	-20	50	z = 170	
c _j - z _j		0	0	0	-20	20-M	-50-M		

